

Catch that butterfly: A Multimodal Approach for Detecting and Simulating Gut Feelings

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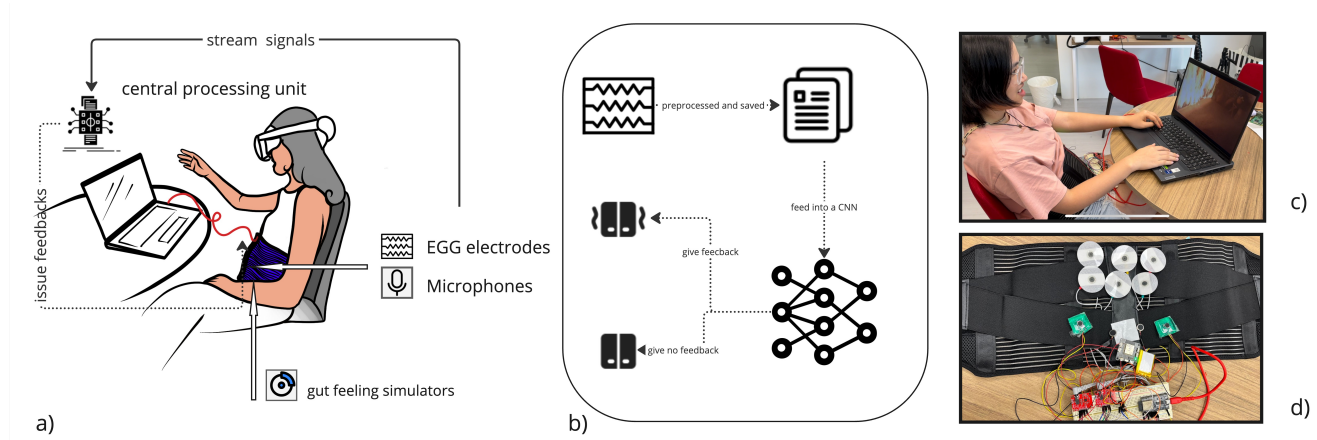


Figure 1: Design of Gutfeel system b) Architecture of the central processing system, c) Data acquisition setup, d) Prototype

Abstract

Gut feelings are an omnipresent sensation underlying many emotional experiences, ranging from positive to negative emotions and even sixth sense intuition. Predicting and stimulating these feelings can enhance and enrich users' emotional experiences. Despite the tremendous potential of gut feelings for HCI design, gut-related signals have been largely ignored in HCI research. In this paper, we introduce an ongoing prototype of an artifact that utilizes both bowel sounds and electrogastrography (EGG) signals to predict and simulate gut feelings. Our prototype consists of an EGG sensing module, a bowel sound sensing module, a machine learning model

to process and predict gut-churning moments, and a feedback mechanism to simulate gut feelings in the abdominal area. We hope this work opens up a new design space for physiological signal-based interaction and offers novel opportunities for enhancing user experiences. For future work, we plan to build a robust dataset for gut feeling prediction and evaluate our artifact's effectiveness in inducing gut feelings in real users.

Keywords

Brain Computer Interface, gut-brain connection, emotion, gut feelings

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1 Introduction

Having butterflies in the stomach? Is it anticipation, or is it anxiety? Is it our sixth sense telling us to stay out of danger, or is it simply hunger? Every waking moment, the gut constantly sends signals to the brain. The gut, scientifically known as the gastrointestinal tract, possesses its own nervous system called the Enteric Nervous System (ENS). This system is estimated to have 100 to 500 million neurons [6], and operates with a significant degree of autonomy. Due to its complexity, the ENS is often referred to as the body's "second brain" or the "gut-brain." Everything that enters our body, apart from the air we breathe, passes through the digestive system. Since the brain's primary role is to regulate the body for survival and reproduction, it takes the information from the ENS very seriously. Upon receiving signals from the ENS, the brain needs to make sense of them [4]. In the process of meaning-making, the brain uses concepts to guide its actions, many of which manifest as emotions [2]. The gut is churning, what does this mean?

When you're watching a sad scene and your gut churns, it's likely sadness that has taken hold. But what about the night before your first overseas trip? That's anticipation! And when your gut churns right before lunchtime? Most likely, it's hunger. Gut feelings are almost omnipresent in our emotional experiences, whether we are aware of it or not. Despite the potential for designing novel interactions with gut feelings, research on this topic within human-computer Interaction (HCI) remains scarce. Most existing works focus on designing sensing methods using EGG [7, 16] and bowel sounds [17, 19]. However, much is not known about how these signals are related to gut feeling moments, how to use them to detect gut feelings, and how to influence user's perception about their guts. In this poster, we present an ongoing prototype of an artifact that can detect "gut feelings" moments from bowel sounds and electrogastrography (EGG) signals and simulate the gut feelings through haptic feedback mechanisms. The potential applications of this artifact include augmenting emotional experiences in immersive environments, enhancing interoceptive awareness, nudging healthy behaviors, and many more.

2 Background

The connection between the gastrointestinal (GI) tract and the brain is often referred to as the "gut-brain axis" [9]. This bidirectional communication system affects various physiological processes and has been increasingly linked to emotional experiences and cognitive functions [1, 12]. Like the heart, the gut has its own pacemaking cells called Interstitial Cells of Cajal (ICC) [13]. These cells send regular electrical impulses, which can be recorded as an electrogastrogram (EGG).

Compared to ECG signals, EGG signals are much slower, with pulses typically recorded at 0.05Hz to 0.5Hz [18], representing gut muscle movements. Prior studies have shown correlations between EGG and arousal [15], EGG and valence [16] when emotions are induced by stimuli. EGG signals also correlate well with feelings of nausea induced by cybersickness [8].

The gastrointestinal system contains gas and liquids. The movements of the gut cause the movements of these substances, creating bowel sounds (also called abdominal sounds). Bowel sounds have been extensively studied in medical science, and listening to gut

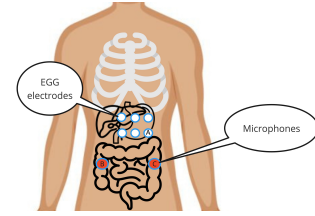


Figure 2: Sensors's positions: A: EGG sensors, below the diaphragm; B, C: microphones

sounds is a common practice for diagnosing problems in the GI tract. For instance, bowel sounds have been used to predict issues in people with irritable bowel syndrome [11]. Recently, bowel sounds have also been used to predict stress [3].

Given that gut-churning often accompanies emotional experiences, such as the sensation of butterflies in the stomach, we explore the use of bowel sounds to capture these sensations.

3 System Design

Our system consists of a sensing system, a central controller, and a feedback system. The sensing system consists of a bowel sound sensing module and an EGG sensing module. For the EGG module, we customized a data acquisition unit based on the open-source design of OpenBCI Cyton to collect EGG signals from six electrodes. Due to the low-frequency nature of the EGG signals, we repurposed the disposable Ag/AgCl pre-gelled electrodes, typically used for ECG, to record EGG signals. These electrodes are placed below the diaphragm, directly on the stomach surface, to sense gastric slow-waves — the rhythmic electrical activity of the stomach muscles. The bowel sound sensing module utilizes two directional microphones (heading inward) connected to a data acquisition device capable of sampling rates up to 200 kHz. Each microphone is embedded in a gel-pad to minimize skin irritation and is adhered to the abdomen using adhesive pads. The microphones are positioned around the colon area to capture bowel sounds, specifically in positions B and C as illustrated in Figure 2.

Signals from both sensing modules are streamed to a central processing unit. Here, the raw signal waveforms are processed and converted into spectrograms. Sound signals are saved at 10-second intervals, while EGG signals are saved every 3 minutes, accommodating their different frequency characteristics.

We employ a Convolutional Neural Network (CNN) as our core model to predict gut-churning moments. This CNN was pre-trained on a dataset of labeled bowel sounds [5]. As sounds and EGG signals have very different frequency ranges (20-2000Hz for sound and 0-0.5 Hz for EGG), we extract the features from these signals separately and fuse them at later stages to produce the final predictions.

The feedback system is undergoing the iterative process to find the best way to induce gut feelings. At the moment, it consists of two actuators placed near the navel, approximately 10 cm apart. These actuators are controlled by a system developed by [10]. To replicate the perception of gut feelings, we create a sensation of a wave running between the actuators by activating them at different frequencies. This feedback is triggered when the controlling system identifies a gut feeling moment. In the current iteration, our feedback system acts as an amplifier of gut feelings, bringing

these subtle signals into the user's conscious awareness to enhance interoception (the perception of internal bodily sensations).

4 Discussion and Future Work

In this work, we introduce an artifact design and a closed-loop pipeline for sensing gut signals, predicting gut feelings, and sending feedback. Our work is a timely response to the call for novel brain-computer interactions during the NeuroCHI 2024 panel discussion [14]. To our knowledge, we are the first to propose the use of bowel sounds in combination with EGG signals to detect gut feelings, as well as making use of gut feelings to fabricate emotional experiences. Our proposal goes beyond affective state prediction and taps into engineering experiences with gut signals.

In the future work, we plan to collect a high quality dataset for gut feelings prediction, build a robust detection and classification model, and experiment different feedback mechanisms (such as electrical muscle stimulation) to induce gut feelings, change user's perceptions in immersive experiences, or improve their interoceptive ability.

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